



NumPy Complete Assessment

NumPy • Student: Sridhar C

⚠️ Auto-Submitted (Tab Switch Detected)

0.0%

Score

0

Points Earned

100

Total Points

100

Total Questions

← Dashboard

Feedback

 **Keep practicing! Review the material and retake the quiz.**

Detailed Answers

Question 1

0 / 1 marks

What is the output?

```
a=np.array([1,2,3])  
b=a.reshape(3,1)  
print(a+b)
```

- A. [2 4 6]
- B. **[[2 3 4],[3 4 5],[4 5 6]]** 
- C. Error
- D. **[[2],[4],[6]]**

Question 2

0 / 1 marks

What is the output?

```
a=np.arange(6).reshape(2,3)  
print(a.T.shape)
```

- A. (2,3)
- B. **(3,2)** 

[← Dashboard](#)

C. (6,)

D. Error

Question 3

0 / 1 marks

What is the output?

```
a=np.array([1,2,3])  
b=a.astype(float)  
print(b.dtype)
```

A. int64

B. float32

C. float64 

D. object

Question 4

0 / 1 marks

What happens?

```
a=np.array([1,2,3])  
b=a.copy()  
b[0]=100  
print(a[0])
```

A. 1 

B. 100

C. Error

[← Dashboard](#)

D. None

Question 5

0 / 1 marks

What is output?

```
a=np.array([[1,2,3],[4,5,6]])  
print(a[:,1])
```

A. [2,5] ☒

B. [[2],[5]]

C. [1,4]

D. Error

Question 6

0 / 1 marks

What is output?

```
a=np.array([[1,2],[3,4]])  
print(a.flatten().shape)
```

A. (2,2)

B. (4,) ☒

C. (1,4)

D. Error

Question 7

← Dashboard

What is output?

```
a=np.array([1,2,3])  
print(np.repeat(a,2))
```

A. [1,2,3,1,2,3]

B. [1,1,2,2,3,3] ✓

C. [2,4,6]

D. Error

Question 8

0 / 1 marks

What is output?

```
a=np.array([1,2,3,4])  
print(np.flip(a))
```

A. [1,2,3,4]

B. [4,3,2,1] ✓

C. [2,3,4,1]

D. Error

Question 9

0 / 1 marks

What is output?

```
a=np.array([1,2,3,4])  
print(np.delete(a,1))
```

A. [1,3,4] ✓

← Dashboard

B. [2,3,4]

C. [1,2,3]

D. Error

Question 10

0 / 1 marks

What is output?

```
a=np.array([1,2,2,3])  
print(np.unique(a))
```

A. [1,2,2,3]

B. [1,2,3] ✓

C. [2,3]

D. Error

Question 11

0 / 1 marks

What is output?

```
a=np.array([[1,2],[3,4]])  
print(a[1,0])
```

A. 1

B. 2

C. 3 ✓

← Dashboard

D. 4

Question 12

0 / 1 marks

What is output?

```
a=np.array([10,20,30,40])  
print(a[-2])
```

A. 20

B. 30 ☒

C. 40

D. 10

Question 13

0 / 1 marks

What is output?

```
a=np.array([0,1,2,3,4])  
print(a[:2])
```

A. [1,3]

B. [0,2,4] ☒

C. [2,4]

D. [0,1,2]

Question 14

← Dashboard

What is output?

```
a=np.array([1,2,3,4])  
print(a[a>2])
```

- A. [1,2]
- B. [3,4] ☒
- C. [True,True]
- D. Error

Question 15

0 / 1 marks

What is output?

```
a=np.array([1,2,3])  
print(a+5)
```

- A. [6,7,8] ☒
- B. [1,2,3,5]
- C. Error
- D. [5,5,5]

Question 16

0 / 1 marks

What is output?

```
a=np.array([1,2,3])  
b=np.array([4,5,6])  
print(a*b)
```

- A. [4,10,18] ☒

← Dashboard

B. [5,7,9]

C. Error

D. [1,2,3]

Question 17

0 / 1 marks

What is output?

```
a=np.array([1,2,3])  
print(np.all(a>0))
```

A. True ☒

B. False

C. None

D. Error

Question 18

0 / 1 marks

What is output?

```
a=np.array([0,0,1])  
print(np.any(a))
```

A. True ☒

B. False

C. Error

← Dashboard

D. None

Question 19

0 / 1 marks

What is output?

```
a=np.array([3,1,2])  
print(np.sort(a))
```

A. [3,1,2]

B. [1,2,3] ☒

C. [2,1,3]

D. Error

Question 20

0 / 1 marks

What is output?

```
a=np.array([3,1,2])  
print(np.argsort(a))
```

A. [0,1,2]

B. [1,2,0] ☒

C. [2,1,0]

D. Error

Question 21

← Dashboard

What is output?

```
a=np.array([1,5,3])  
print(np.argmax(a))
```

A. 1 ☒

B. 5

C. 2

D. 0

Question 22

0 / 1 marks

What is output?

```
a=np.array([1,5,3])  
print(np.argmin(a))
```

A. 0 ☒

B. 1

C. 2

D. 3

Question 23

0 / 1 marks

What is output?

```
a=np.array([1,2,3])  
print(np.sum(a))
```

A. 6 ☒

[← Dashboard](#)

B. 5

C. 3

D. Error

Question 24

0 / 1 marks

What is output?

```
a=np.array([1,2,3])  
print(np.mean(a))
```

A. 2 ☒

B. 3

C. 1

D. Error

Question 25

0 / 1 marks

What is output?

```
a=np.array([1,2,3])  
print(np.std(a))
```

A. 0.81 ☒

B. 1

C. 2

← Dashboard

D. Error

Question 26

0 / 1 marks

What is the output?

```
a=np.arange(6).reshape(2,3)
b=a[:,1]
b[0]=100
print(a)
```

A. `[[0 1 2] [3 4 5]]`

B. `[[0 100 2] [3 4 5]]`

C. `[[0 100 2] [3 100 5]]` ✓

D. Error

Question 27

0 / 1 marks

What is the output?

```
a=np.array([[1,2],[3,4]])
b=a.flatten()
b[0]=99
print(a[0,0])
```

A. 1 ✓

B. 99

C. Error

D. None

← Dashboard

Question 28

0 / 1 marks

What is the output?

```
a=np.array([[1,2],[3,4]])  
b=a.ravel()  
b[0]=99  
print(a)
```

- A. `[[1 2] [3 4]]`
- B. `[[99 2] [3 4]]` ☒
- C. Error
- D. None

Question 29

0 / 1 marks

What is the output?

```
a=np.arange(6).reshape(2,3)  
print(a[:,::-1])
```

- A. `[[2 1 0] [5 4 3]]` ☒
- B. `[[0 1 2] [3 4 5]]`
- C. `[[3 4 5] [0 1 2]]`
- D. Error

Question 30

0 / 1 marks

What is the output?

[← Dashboard](#)

```
a=np.array([1,2,3,4])  
b=a[::2]  
b[1]=100  
print(a)
```

A. [1 2 3 4]

B. [1 2 100 4] 

C. [1 100 3 4]

D. Error

Question 31

0 / 1 marks

What is the output?

```
a=np.array([[1,2],[3,4]])  
b=a.T  
b[0,1]=100  
print(a)
```

A. [[1 2] [3 4]]

B. [[1 100] [3 4]]

C. [[1 2] [100 4]] 

D. Error

Question 32

0 / 1 marks

What is the output?

```
a=np.arange(6)  
b=a.reshape(2,3)  
c=b.flatten()
```

[← Dashboard](#)

```
c[0]=99  
print(a[0])
```

A. 0 ☒

B. 99

C. Error

D. None

Question 33

0 / 1 marks

What is the output?

```
a=np.arange(6)  
b=a.reshape(2,3)  
c=b.ravel()  
c[0]=99  
print(a[0])
```

A. 0

B. 99 ☒

C. Error

D. None

Question 34

0 / 1 marks

What is the output?

```
a=np.array([1,2,3])  
b=np.array([[1],[2],[3]])  
print((a+b).shape)
```

← Dashboard

A. (3,)

B. (3,3) ☒

C. (1,3)

D. Error

Question 35

0 / 1 marks

What is the output?

```
a=np.arange(6).reshape(2,3)
print(np.sum(a,axis=1))
```

A. [3 12] ☒

B. [5 7]

C. [6 9]

D. Error

Question 36

0 / 1 marks

What is the output?

```
a=np.arange(6).reshape(2,3)
print(np.sum(a,axis=0))
```

A. [3 5 7] ☒

B. [6 9]

C. [3 12]

[← Dashboard](#)

D. Error

Question 37

0 / 1 marks

What is the output?

```
a=np.array([[1,2],[3,4]])  
print(np.max(a,axis=1))
```

A. [2 4] ☒

B. [3 4]

C. [4]

D. Error

Question 38

0 / 1 marks

What is the output?

```
a=np.array([[1,2],[3,4]])  
print(np.min(a,axis=0))
```

A. [1 2] ☒

B. [2 4]

C. [1 3]

D. Error

Question 39

← Dashboard

What is the output?

```
a=np.array([1,2,3,4])  
print(np.where(a%2==0,a,-1))
```

A. [1 2 3 4]

B. [-1 2 -1 4] 

C. [2 4]

D. Error

Question 40

0 / 1 marks

What is the output?

```
a=np.array([3,1,2])  
b=np.argsort(a)  
print(a[b])
```

A. [3 1 2]

B. [1 2 3] 

C. [2 1 3]

D. Error

Question 41

0 / 1 marks

What is the output?

```
a=np.array([1,2,2,3])  
u,c=np.unique(a,return_counts=True)  
print(u[c>1])
```

[← Dashboard](#)

A. [1 2 3]

B. [2] 

C. [1]

D. Error

Question 42

0 / 1 marks

What is the output?

```
a=np.arange(6).reshape(2,3)
b=np.flip(a)
print(b)
```

A. `[[5 4 3] [2 1 0]]` 

B. `[[2 1 0] [5 4 3]]`

C. `[[0 1 2] [3 4 5]]`

D. Error

Question 43

0 / 1 marks

What is the output?

```
a=np.array([1,2,3])
b=np.repeat(a,2)
print(b.shape)
```

A. (3,)

B. (6,) 

[← Dashboard](#)

C. (2,3)

D. Error

Question 44

0 / 1 marks

What is the output?

```
a=np.arange(6)
b=a.reshape(2,3)
print(b[1,2])
```

A. 5 ☒

B. 4

C. 3

D. Error

Question 45

0 / 1 marks

What is the output?

```
a=np.array([[1,2,3],[4,5,6]])
print(a[0:2,1:3].shape)
```

A. (2,2) ☒

B. (1,2)

C. (2,3)

D. Error

[← Dashboard](#)

Question 46

0 / 1 marks

What is the output?

```
a=np.array([1,2,3])  
b=np.array([4,5,6])  
print(np.dot(a,b))
```

A. [4 10 18]

B. 32 ☒

C. 15

D. Error

Question 47

0 / 1 marks

What is the output?

```
a=np.array([1,4,9])  
print(np.sqrt(a))
```

A. [1 2 3]

B. [1. 2. 3.] ☒

C. [1 4 9]

D. Error

Question 48

0 / 1 marks

What is the output?

[← Dashboard](#)

```
a=np.array([1.234,2.567])  
print(np.round(a,1))
```

A. [1.2 2.6] ☒

B. [1.23 2.57]

C. [1 2]

D. Error

Question 49

0 / 1 marks

What is the output?

```
a=np.array([1.7,2.3])  
print(np.floor(a))
```

A. [2 3]

B. [1. 2.] ☒

C. [1.7 2.3]

D. Error

Question 50

0 / 1 marks

What is the output?

```
a=np.array([1,2,3])  
b=a.astype(float)  
print(b*2)
```

A. [2 4 6]

[← Dashboard](#)

B. [2. 4. 6.] ☒

C. Error

D. [1 2 3]

Question 51

0 / 1 marks

What is the output?

```
a=np.arange(6).reshape(2,3)
b=a[:,1:2]
b[0,0]=99
print(a)
```

A. `[[0 1 2] [3 4 5]]`

B. `[[0 99 2] [3 4 5]]` ☒

C. `[[0 99 2] [3 99 5]]`

D. Error

Question 52

0 / 1 marks

What is the output?

```
a=np.array([[1,2],[3,4]])
b=a[:,[1]]
b[0,0]=99
print(a)
```

A. `[[1 2] [3 4]]` ☒

B. `[[1 99] [3 4]]`

[← Dashboard](#)

C. `[[1 99] [3 99]]`

D. Error

Question 53

0 / 1 marks

What is the output?

```
a=np.arange(6)
b=a.reshape(2,3)
c=b.T
c[0,1]=99
print(a)
```

A. `[0 1 2 3 4 5]`

B. `[0 1 99 3 4 5]` ✓

C. `[0 99 2 3 4 5]`

D. Error

Question 54

0 / 1 marks

What is the output?

```
a=np.arange(6).reshape(2,3)
b=a.copy()
b[0,0]=99
print(a[0,0])
```

A. 0 ✓

B. 99

C. Error

← Dashboard

D. None

Question 55

0 / 1 marks

What is the output?

```
a=np.array([1,2,3,4])  
b=a[[0,2]]  
b[1]=99  
print(a)
```

A. [1 2 99 4]

B. [1 2 3 4] ☒

C. [99 2 3 4]

D. Error

Question 56

0 / 1 marks

What is the output?

```
a=np.array([[1,2],[3,4]])  
print(np.sum(a,axis=1,keepdims=True))
```

A. [[3 7]]

B. [[3],[7]] ☒

C. [3 7]

D. Error

[← Dashboard](#)

Question 57

0 / 1 marks

What is the output?

```
a=np.arange(4).reshape(2,2)
b=np.arange(2)
print(a+b)
```

A. `[[0 2] [2 4]]` ✓

B. `[[0 1] [2 3]]`

C. Error

D. `[[1 2] [3 4]]`

Question 58

0 / 1 marks

What is the output?

```
a=np.array([[1,2,3]])
b=np.array([[1],[2],[3]])
print(a*b)
```

A. Error

B. 3x3 matrix ✓

C. 1x3 matrix

D. 3x1 matrix

Question 59

0 / 1 marks

What is the output?

← Dashboard

```
a=np.arange(6).reshape(2,3)
print(a.reshape(-1,2))
```

A. (3,2) ☒

B. (2,3)

C. Error

D. (6,1)

Question 60

0 / 1 marks

What is the output?

```
a=np.array([1,2,3])
print(np.expand_dims(a,axis=0).shape)
```

A. (3,)

B. (1,3) ☒

C. (3,1)

D. Error

Question 61

0 / 1 marks

What is the output?

```
a=np.array([1,2,3])
print(np.expand_dims(a,axis=1).shape)
```

A. (3,)

B. (1,3)

[← Dashboard](#)

C. (3,1) ☒

D. Error

Question 62

0 / 1 marks

What is the output?

```
a=np.array([[1,2],[3,4]])  
print(np.squeeze(a).shape)
```

A. (2,2) ☒

B. (4,)

C. (2,)

D. Error

Question 63

0 / 1 marks

What is the output?

```
a=np.array([1,2,3])  
b=np.array([1,2])  
print(a+b)
```

A. [2 4 6]

B. Error ☒

C. [2 4]

D. Broadcasts partially

← Dashboard

Question 64

0 / 1 marks

What is the output?

```
a=np.arange(6).reshape(2,3)
print(a[:,[1,1]])
```

A. `[[1 1] [4 4]]` 

B. `[[1 4] [1 4]]`

C. `[[1 1] [1 1]]`

D. Error

Question 65

0 / 1 marks

What is the output?

```
a=np.arange(6).reshape(2,3)
print(a[[0,1],[1,2]])
```

A. `[[1 2] [4 5]]`

B. `[1 5]` 

C. `[4 2]`

D. Error

Question 66

0 / 1 marks

What is the output?

```
a=np.array([1,2,3])
```

[← Dashboard](#)

```
b=np.array([[1],[2]])  
print((a+b).shape)
```

- A. (3,)
- B. (2,3) ☒
- C. (3,2)
- D. Error

Question 67

0 / 1 marks

What is the output?

```
a=np.array([[1,2],[3,4]])  
print(np.mean(a,axis=0))
```

- A. [2 3]
- B. [2. 3.]
- C. [1.5 3.5] ☒
- D. Error

Question 68

0 / 1 marks

What is the output?

```
a=np.array([[1,2],[3,4]])  
print(np.mean(a,axis=1))
```

- A. [1.5 3.5] ☒
- B. [2 3]

[← Dashboard](#)

C. [3 7]

D. Error

Question 69

0 / 1 marks

What is the output?

```
a=np.array([1,2,3])  
print(np.var(a))
```

A. 1

B. 0.66

C. 0.6666 

D. Error

Question 70

0 / 1 marks

What is the output?

```
a=np.array([1,2,3])  
print(np.std(a))
```

A. 1

B. 0.816 

C. 0.666

D. Error

[← Dashboard](#)

Question 71

0 / 1 marks

What is the output?

```
a=np.array([1,2,3])  
print(np.prod(a))
```

A. 6 ☒

B. 5

C. 3

D. Error

Question 72

0 / 1 marks

What is the output?

```
a=np.array([1,4,9])  
print(np.sqrt(a).dtype)
```

A. int64

B. float64 ☒

C. object

D. bool

Question 73

0 / 1 marks

What is the output?

```
a=np.array([1.9,2.1])  
print(np.floor(a))
```

← Dashboard

A. [2 3]

B. [1. 2.] 

C. [1 2]

D. Error

Question 74

0 / 1 marks

What is the output?

```
a=np.array([1,2,3])  
print(np.power(a,2))
```

A. [2 4 6]

B. [1 4 9] 

C. [1 2 3 2]

D. Error

Question 75

0 / 1 marks

What is the output?

```
a=np.array([1,2,3])  
b=np.array([0,1,0])  
print(np.where(b,a,-1))
```

A. [1 2 3]

B. [-1 2 -1] 

C. [1 -1 3]

[← Dashboard](#)

D. Error

Question 76

0 / 1 marks

What does `np.array([1,2,3])` create?

A. A Python list

B. A 1D NumPy array ☒

C. A tuple

D. A set

Question 77

0 / 1 marks

What is the shape of `np.array([[1,2,3],[4,5,6]])`?

A. (6,)

B. (3,2)

C. (2,3) ☒

D. (1,6)

Question 78

0 / 1 marks

What does `arr.size` return?

A. Number of rows

B. Number of columns

[← Dashboard](#)

C. Total elements ☒

D. Shape

Question 79

0 / 1 marks

What does `arr.ndim` represent?

A. Number of elements

B. Number of dimensions ☒

C. Data type

D. Size

Question 80

0 / 1 marks

What is output of `np.arange(5)`?

A. [1 2 3 4 5]

B. [0 1 2 3 4] ☒

C. [0 1 2 3 4 5]

D. [1 2 3 4]

Question 81

0 / 1 marks

How many elements in `np.linspace(0,10,5)`?

A. 4

← Dashboard

B. 5 ☒

C. 10

D. 11

Question 82

0 / 1 marks

What does `np.zeros(3)` return?

A. [1 1 1]

B. [0 0 0]

C. [0. 0. 0.] ☒

D. Empty

Question 83

0 / 1 marks

What does `np.ones(3)` return?

A. [1. 1. 1.] ☒

B. [1 1 1]

C. [0 0 0]

D. Error

Question 84

What does `np.full(3,7)` return?

[← Dashboard](#)

A. [7 7 7] ☒

B. [3 3 3]

C. [0 0 0]

D. Error

Question 85

0 / 1 marks

What does `np.identity(2)` return?

A. `[[1 1],[1 1]]`

B. `[[1 0],[0 1]]` ☒

C. `[[0 0],[0 0]]`

D. Error

Question 86

0 / 1 marks

What does `arr.flatten()` do?

A. Returns 2D

B. Returns 1D copy ☒

C. Returns view

D. Deletes data

[← Dashboard](#)

Question 87

0 / 1 marks

What does arr.T do?

A. Transpose array ☒

B. Flatten array

C. Sort array

D. Delete rows

Question 88

0 / 1 marks

What does np.flip([1,2,3]) return?

A. [1 2 3]

B. [3 2 1] ☒

C. [2 3 1]

D. Error

Question 89

0 / 1 marks

What does np.concatenate do?

A. Splits arrays

B. Joins arrays ☒

C. Sorts arrays

D. Deletes elements

← Dashboard

Question 90

0 / 1 marks

What does `np.repeat([1,2],2)` return?

- A. [1 2 1 2]
- B. [1 1 2 2] ☒
- C. [2 4]
- D. Error

Question 91

0 / 1 marks

What does `np.unique([1,2,2,3])` return?

- A. [1 2 2 3]
- B. [1 2 3] ☒
- C. [2 3]
- D. Error

Question 92

0 / 1 marks

What does `np.random.randint(0,5,3)` generate?

- A. 3 floats
- B. 3 ints between 0-4 ☒
- C. 5 ints

[← Dashboard](#)

D. Error

Question 93

0 / 1 marks

What range does `np.random.rand()` follow?

A. `[0,1)` ✓

B. `[1,10)`

C. `[-1,1)`

D. `[0,100)`

Question 94

0 / 1 marks

What does `np.all([True,True])` return?

A. `True` ✓

B. `False`

C. `None`

D. `Error`

Question 95

0 / 1 marks

What does `np.any([False,True])` return?

A. `False`

B. `True` ✓

← Dashboard

C. None

D. Error

Question 96

0 / 1 marks

What does `np.sort([3,1,2])` return?

A. `[3 1 2]`

B. `[1 2 3]` ✓

C. `[2 1 3]`

D. Error

Question 97

0 / 1 marks

What does `np.max([1,5,3])` return?

A. 1

B. 5 ✓

C. 3

D. Error

Question 98

0 / 1 marks

What does `np.min([1,5,3])` return?

A. 1 ✓

← Dashboard

B. 5

C. 3

D. Error

Question 99

0 / 1 marks

What does `np.sum([1,2,3])` return?

A. 6 ☒

B. 5

C. 3

D. Error

Question 100

0 / 1 marks

What does `np.mean([1,2,3])` return?

A. 2 ☒

B. 3

C. 1

D. Error